

Alireza Azizi

Ph.D. Student, Department of Electrical and Computer Engineering, Rice University
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Rice University | Google Scholar

RESEARCH INTERESTS

Theoretical computer science, learning theory, distribution testing, prediction-augmented algorithms, information theory, signal processing, computational imaging, and wireless sensing.

EDUCATION

Rice University Houston, TX, USA
Ph.D. in Electrical and Computer Engineering Aug. 2022–Present
GPA: 4.00/4.00

Rice University Houston, TX, USA
M.S. in Electrical and Computer Engineering degree to be conferred Aug. 2026
GPA: 4.00/4.00

Sharif University of Technology Tehran, Iran
B.S. in Electrical Engineering Sept. 2015–Jan. 2020
GPA: 18.81/20 (3.96/4.00)

Sharif University of Technology Tehran, Iran
Graduate study in Electrical Engineering Sept. 2020–2022
GPA: 18.51/20 (4.00/4.00); withdrew upon joining Rice University.

RESEARCH APPOINTMENTS AND VISITS

Simons Institute for the Theory of Computing, UC Berkeley Berkeley, CA, USA
Visiting Graduate Student Researcher, Federated and Collaborative Learning Program Spring 2026

- Participating in a semester-long research program on the theoretical foundations of federated and collaborative learning, including reliable, efficient, privacy-preserving, and trustworthy learning in distributed and data-sensitive environments.
- Simons profile: simons.berkeley.edu/people/alireza-azizi.

PUBLICATIONS AND MANUSCRIPTS

1. M. Aliakbarpour, **A. Azizi**, and R. Stevens. “Optimal Prediction-Augmented Algorithms for Testing Independence of Distributions.” *Conference on Learning Theory (COLT)*. 2026. **Accepted** for publication. Also available as arXiv:2603.04635.
2. A. Deshwal, **A. Azizi**, S. Sinha, D. Pandey, N. Mehrotra, A. Pal, and A. Sabharwal. “Multi Person Localization and Breathing Rate Estimation using Translational and Rotational Synthetic Aperture Radar.” *IEEE Transactions on Computational Imaging*. **Accepted** for publication. Not yet indexed on Google Scholar.
3. S. Sinha, A. Deshwal, **A. Azizi**, D. Pandey, N. Mehrotra, A. Pal, and A. Sabharwal. “Through-the-Wall Multi-Person Localization using Translation and Rotation Synthetic Aperture Radar.” *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. 2025. [IEEE Xplore](#).
4. L. Khaloopour, S. V. Rouzegar, **A. Azizi**, A. Hosseinian, M. Farahnak-Ghazani, M. Nasiri-Kenari, and M. Mir-mohseni. “An Experimental Platform for Macro-Scale Fluidic Medium Molecular Communication.” *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*. 2020. [IEEE Xplore](#).

RESEARCH EXPERIENCE

Graduate Researcher, Rice University Houston, TX, USA
Learning Theory, Prediction-Augmented Algorithms, and Optimization 2025–Present

- Develop sample-efficient prediction-augmented algorithms for distribution testing, including independence testing of high-dimensional discrete distributions with auxiliary prediction distributions.
- Contributed to a COLT 2026 paper on optimal prediction-augmented algorithms for independence testing, with robustness guarantees under imperfect predictions.

- Pursue ongoing research directions in augmented algorithms for entropy estimation, heterogeneous optimization, and augmented algorithms for private mean estimation.
- Analyze sample-complexity guarantees using tools from distribution testing, total variation distance, Poissonization, information-theoretic lower bounds, and concentration analysis.

Graduate Researcher, Rice University
Computational Imaging and Wireless Sensing

Houston, TX, USA
 Aug. 2022–Aug. 2025

- Designed signal processing and synthetic-aperture radar methods for through-wall and non-line-of-sight human localization and breathing-rate estimation.
- Worked with UWB and mmWave radar data to localize multiple live targets through scattering media and validate detections using respiration signatures.
- Contributed to publications in ICASSP and IEEE Transactions on Computational Imaging.

Undergraduate Researcher, Sharif University of Technology
Macro-Scale Molecular Communication Experimental Platform

Tehran, Iran
 Aug. 2018–Oct. 2019

- Designed and built a macro-scale fluidic molecular communication platform using a controlled transmitter, pH-meter receiver, and microcontroller-based data acquisition.
- Collected channel-response and error-rate measurements and contributed to the experimental results of a journal publication.

HONORS AND AWARDS

- Ranked **10th** among approximately 10,000 participants in Iran’s Electrical Engineering Entrance Exam for Master’s degree.
- Ranked **74th** among 167,868 participants in Iran’s National University Entrance Exam, 2015.

SELECTED GRADUATE COURSEWORK

Theory of Machine Learning; Information Theory and Coding; Introduction to Random Processes and Applications; Linear Algebra for Data Science; Algorithms, Complexity, and Approximation; Network Science and Analytics; Probabilistic Toolkit for Learning and Computation; Optimization Theory; Advanced Digital Signal Processing; Blind Source Separation; Network Coding and Applications.

TEACHING EXPERIENCE

Teaching Assistant: Machine Learning for Data Science (2024, 2025); Signals and Systems (Spring 2017); Digital Signal Processing (Spring 2021); Engineering Mathematics (Fall 2020, Fall 2021); Circuit Theory (Spring 2018, Spring 2020, Spring 2022); Numerical Computation (Spring 2017, Fall 2018); EE Principles Lab (Fall 2017); Electronics Principles Lab (Fall 2018).

TECHNICAL SKILLS

Programming and scientific computing	MATLAB, Python, C, C++
Typesetting and research tools	LaTeX, Overleaf, Google Scholar, IEEE Xplore

LANGUAGES

- English: Fluent.
- Persian: Native.

Last updated: May 2026